Hevo Assesment

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2. `README.md`

Hevo Assessment I: Data Pipeline Setup & Transformation

2.1. Project Overview

- **Goal:** Set up a CDC-based data pipeline from a local PostgreSQL source to a Snowflake destination using Hevo Data, including data transformation.

2.2. Prerequisites & Accounts Used

- **Hevo Account Details:**
  - **Team Name:** siddharthkakade1999@gmail.com
  - **Pipeline ID: #7**
- **Local Postgres Setup:** Docker desktop running latest postgres image

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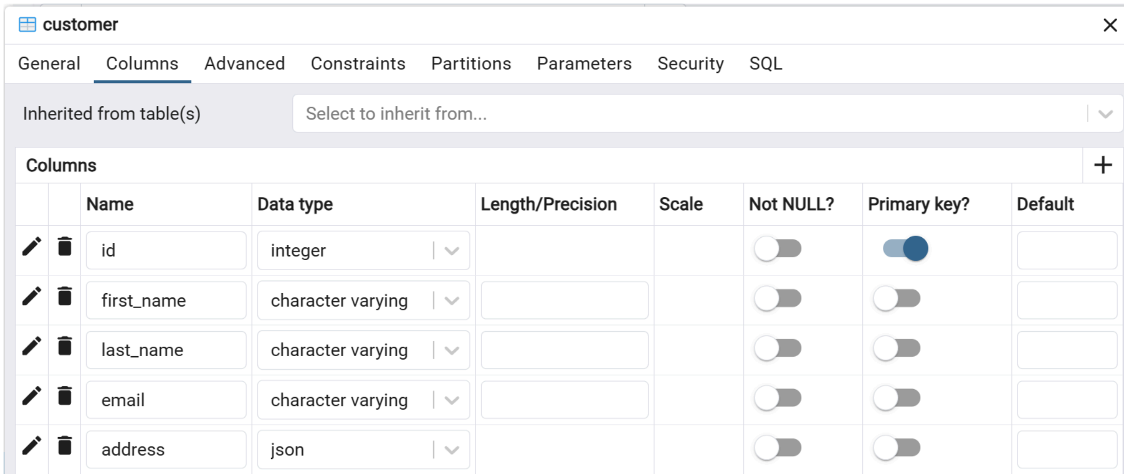
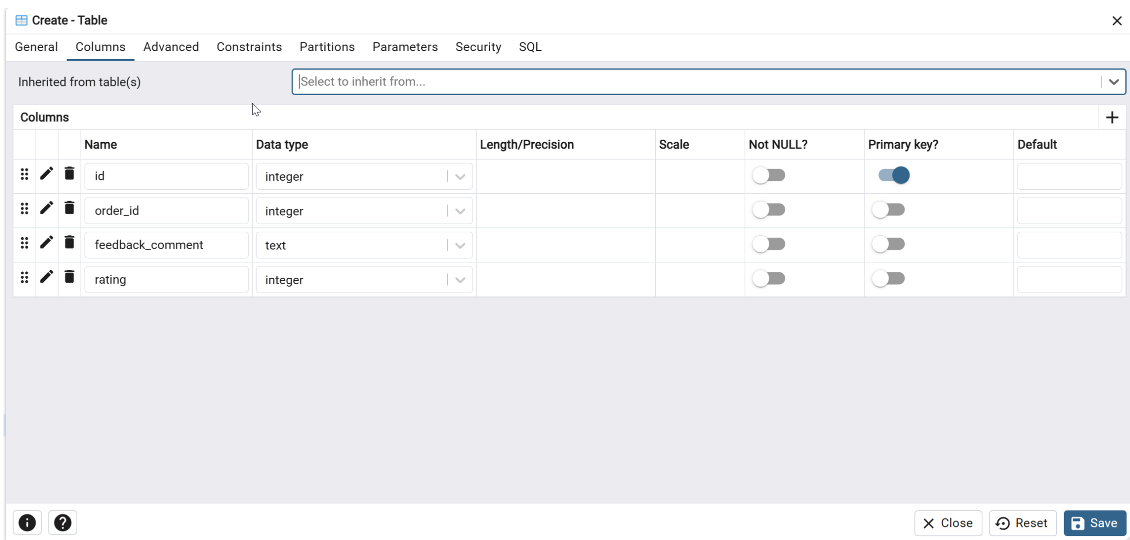
2.3. Detailed Steps to Reproduce the Setup

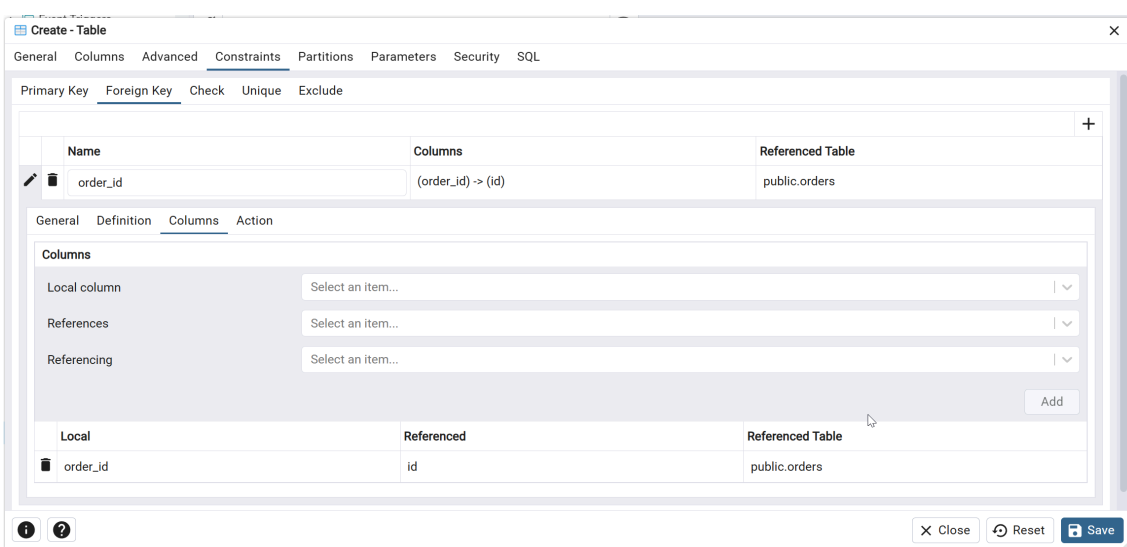
2.3.1. Database Setup (PostgreSQL)

1. **Installation:**
   Command ran in docker desktop:
   `docker run --name mydatabase2 -e POSTGRES_PASSWORD=admin -e POSTGRES_USER=admin -e POSTGRES_DB=mydatabase2 -p 5433:5432 -d postgres`
2. **Schema Creation:**
   a. Customer table:

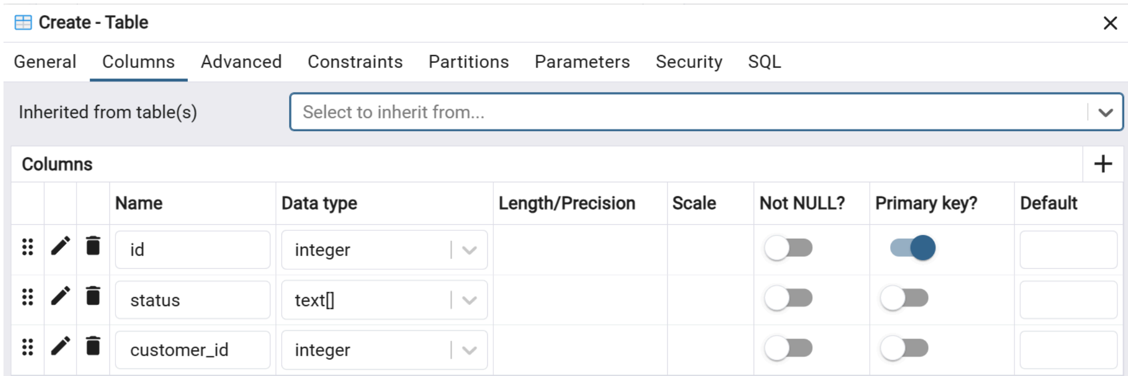


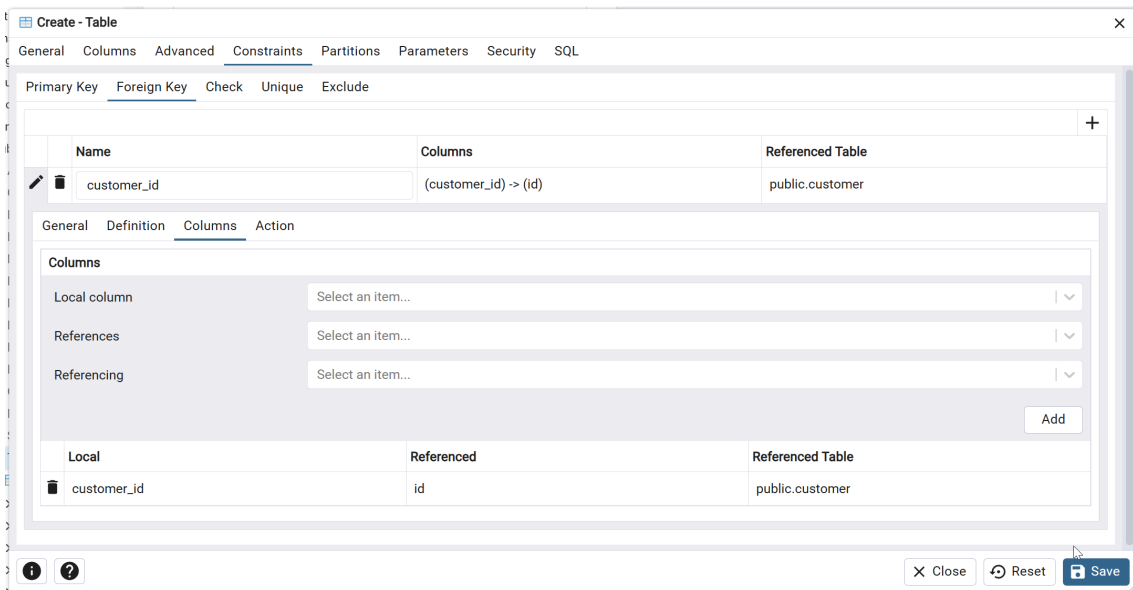
b. Feedback table:





c. Order table:





3. **Data Loading:** pg4admin along with CSV available on [GitHub - muskan-kesharwani-hevo/hevo-assessment-csv: CSV files for Hevo interview assessment]()

   a. Further CSV was imported by following - https://www.youtube.com/watch?v=Ikd2xSb00UI

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4. **Networking/Connectivity:**

- Command to enter the docker container to run the ngrok util: `docker exec -it mydatabase /bin/bas`
- Using ngrok command: `ngrok tcp 5432`
- Further `wal_level` was set to 10
- Restart agent service and granted the permission along with whitelsiting the URL

2.3.2. Hevo Pipeline Setup (Objective 5)

1. **Source Configuration:**
    - Type: PostgreSQL
    - **Ingestion Mode: Logical Replication**
    - further set `wal_level = logical`
2. **Destination Configuration:**
    - Type: Snowflake Partner Connect
    - Schema Name: `PC_HEVODATA_DB`

2.4. Transformation Details (Objective 6)

2.4.1. Transformation 1:

- **Goal:** To create a new table in Snowflake as order_events where each row shows as `event_type = 'order_delivered'.`
- **Method Used:** Python
- **Reference**: https://docs.hevodata.com/pipelines/transformations/python-transfm/utils/retrieve-the-type-for-value/
- **Code**:
    - Example

```
from io.hevo.api import Event
from io.hevo.api import Utils
```

```
def transform(event):
    properties = event.getProperties()
    order_status = properties['status']

    event_properties = {
        'order_id': properties.get('order_id')
    }

    event_type = None
    if order_status == 'placed':
        event_type = 'order_placed'
    elif order_status == 'shipped':
        event_type = 'order_shipped'
    elif order_status == 'delivered':
        event_type = 'order_delivered'
    elif order_status == 'cancelled':
        event_type = 'order_cancelled'

    if event_type:
        event_properties['event_type'] = event_type
        new_event = Event('order_events', event_properties)
        return [new_event]

    return event
```

2.4.2. Transformation 2:

- **Goal:** Extract the text before the `@` symbol from the `email` field as username
- **Method Used:** Python
- **Code**:
    - Sample

```
from io.hevo.api import Event
from io.hevo.api import Utils


def transform(event):
    properties = event.getProperties()

    email_field = properties.get('email')
    if isinstance(email_field, str) and '@' in email_field:
        username = email_field.split('@', 1)[0]
        properties['username'] = username
        event.setProperties(properties)
    return event
```

- **Example:** `jane.doe@gmail.com` creates an table with username `jane.doe`

2.5. Assumptions Made & Choices

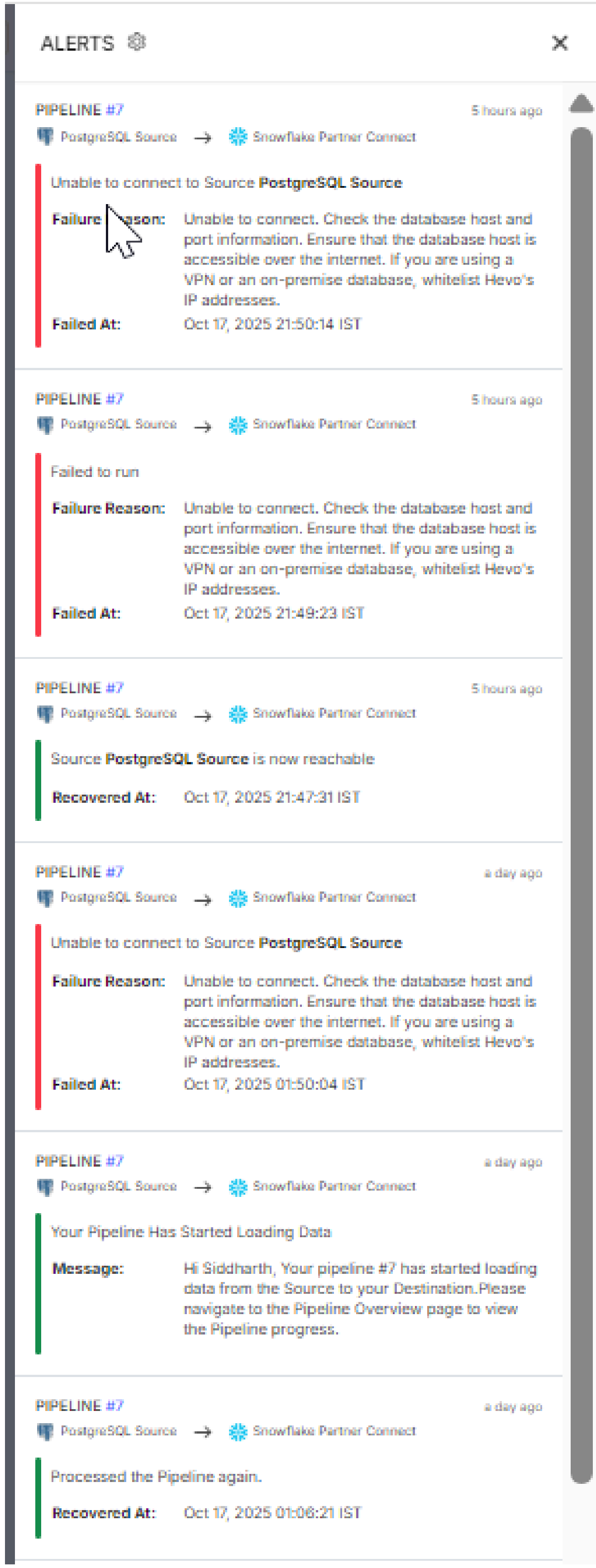
- **`status` Field:** *Assumption:* The `status` field in the Postgres DDL was created as a `VARCHAR` instead of a custom Postgres `ENUM` for simplicity/wider compatibility.)
- **Transformation Language:** Python

2.6. Issues, Roadblocks, & Workarounds

- The main issue rev**olved around** set**ting** up the postgres set up I'm connecting the local database to the Hevo pipeline.
- The local database was not able to connect to the Hevo pipeline and to troubleshoot this we restarted the container restarted the service configured necessary as per the support article but still there were few network issues
- Using psql found out where are the necessary file to edit as prerequisite. Configured files `postgresql.conf` and `pg_hba.conf` by searching via psql `SHOW hba_file;` and `SHOW config_file;`
- Few times the database was able to connect to the servers using ngrok but still either due to high latency or some kind of network issues the further

connection was not established

- Multiple network issue addressed during the testing


ALERTS
PIPELINE #7
5 hours ago
PostgreSQL Source → Snowflake Partner Connect
Unable to connect to Source **PostgreSQL Source**
Failure **ason:**
Unable to connect. Check the database host and port information. Ensure that the database host is accessible over the internet. If you are using a VPN or an on-premise database, whitelist Hevo's IP addresses.
Failed At:
Oct 17, 2025 21:50:14 IST
PIPELINE #7
5 hours ago
PostgreSQL Source → Snowflake Partner Connect
Failed to run
Failure Reason:
Unable to connect. Check the database host and port information. Ensure that the database host is accessible over the internet. If you are using a VPN or an on-premise database, whitelist Hevo's IP addresses.
Failed At:
Oct 17, 2025 21:49:23 IST
PIPELINE #7
5 hours ago
PostgreSQL Source → Snowflake Partner Connect
Source **PostgreSQL Source** is now reachable
Recovered At:
Oct 17, 2025 21:47:31 IST
PIPELINE #7
a day ago
PostgreSQL Source → Snowflake Partner Connect
Unable to connect to Source **PostgreSQL Source**
Failure Reason:
Unable to connect. Check the database host and port information. Ensure that the database host is accessible over the internet. If you are using a VPN or an on-premise database, whitelist Hevo's IP addresses.
Failed At:
Oct 17, 2025 01:50:04 IST
PIPELINE #7
a day ago
PostgreSQL Source → Snowflake Partner Connect
Your Pipeline Has Started Loading Data
Message:
Hi Siddharth, Your pipeline #7 has started loading data from the Source to your Destination.Please navigate to the Pipeline Overview page to view the Pipeline progress.
PIPELINE #7
a day ago
PostgreSQL Source → Snowflake Partner Connect
Processed the Pipeline again.
Recovered At:
Oct 17, 2025 01:06:21 IST

2.7. **Validation Check on Snowflake:**